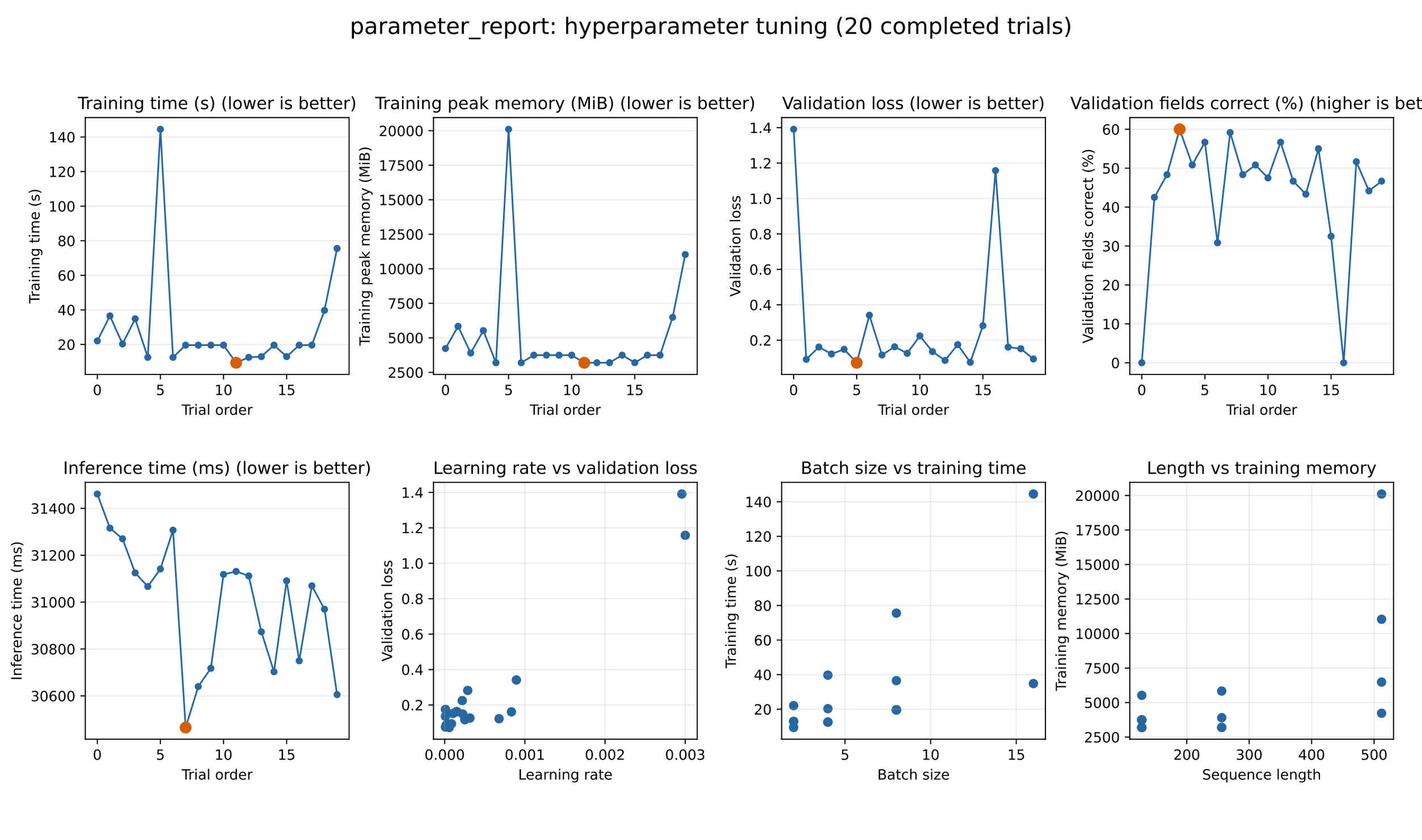
Search environments

parameter_report
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

parameter_report_flash (flash_attention)
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

parameter_report_kvcache (kvcache)
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

parameter_report_lora (lora)
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python


parameter_report: hyperparameter tuning (20 completed trials)
Training time (s) (lower is better)
Training time (s)
Trial order
Training peak memory (MiB) (lower is better)
Training peak memory (MiB)
Trial order
Validation loss (lower is better)
Validation loss
Trial order
Validation fields correct (%) (higher is bet
Validation fields correct (%)
Trial order
Inference time (ms) (lower is better)
Inference time (ms)
Trial order
Learning rate vs validation loss
Validation loss
Learning rate
Batch size vs training time
Training time (s)
Batch size
Length vs training memory
Training memory (MiB)
Sequence length

parameter_report: trial hyperparameters

Trial	Learning Rate	Batch Size	Length
0	0.0029578	2	512
1	3.5613e-05	8	256
2	0.00015536	4	256
3	0.00067814	16	128
4	0.00022618	4	128
5	5.6412e-05	16	512
6	0.00089414	4	128
7	**0.00025198**	**8**	**128**
8	0.00014895	8	128
9	0.00031556	8	128
10	0.00021928	8	128
11	1e-05	2	128
12	1.8304e-05	4	128
13	1e-05	2	256
14	1e-05	8	128
15	0.0002873	2	256
16	0.003	8	128
17	0.00083278	8	128
18	0.00010385	4	512
19	8.5512e-05	8	512

Highlighted row: selected best trial based on this report's objectives.

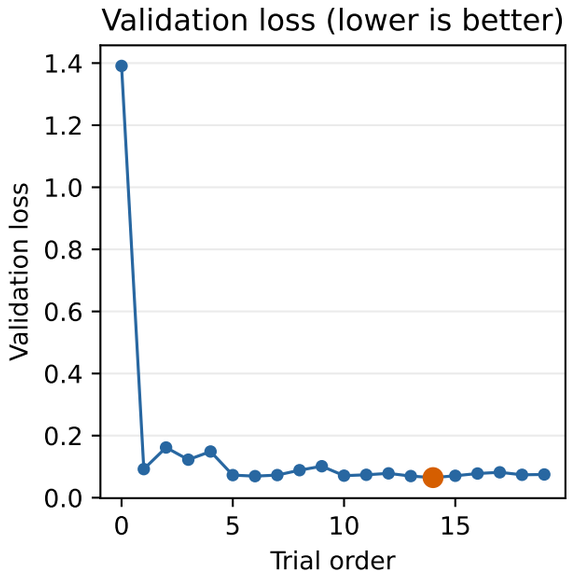
parameter_report_flash (flash_attention)

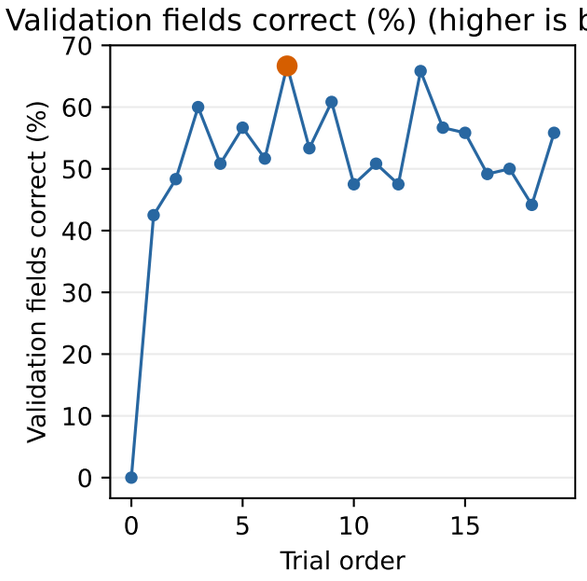
No completed trials were found.

parameter_report_flash (flash_attention): trial hyperparameters

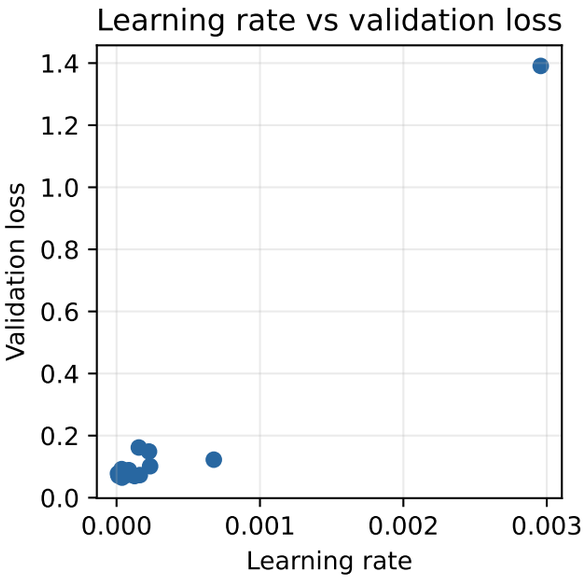
No completed trials were found.

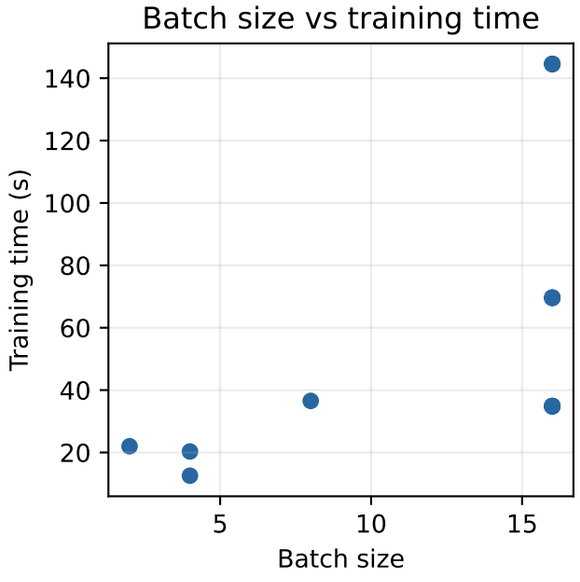
parameter_report_kvcache (kvcache): hyperparameter tuning (20 completed trials)

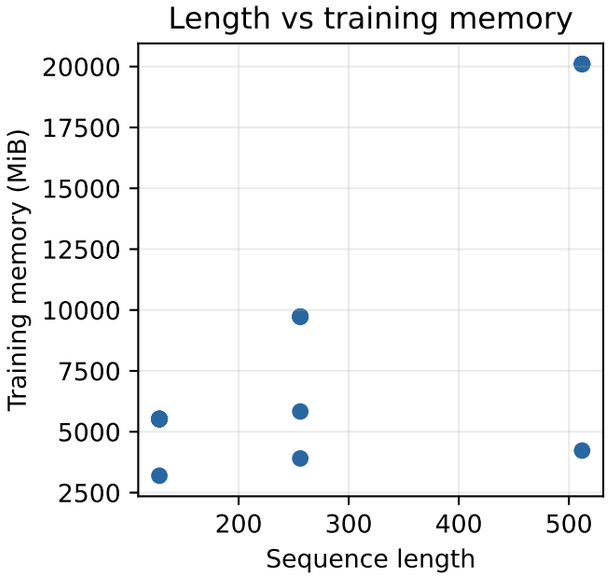








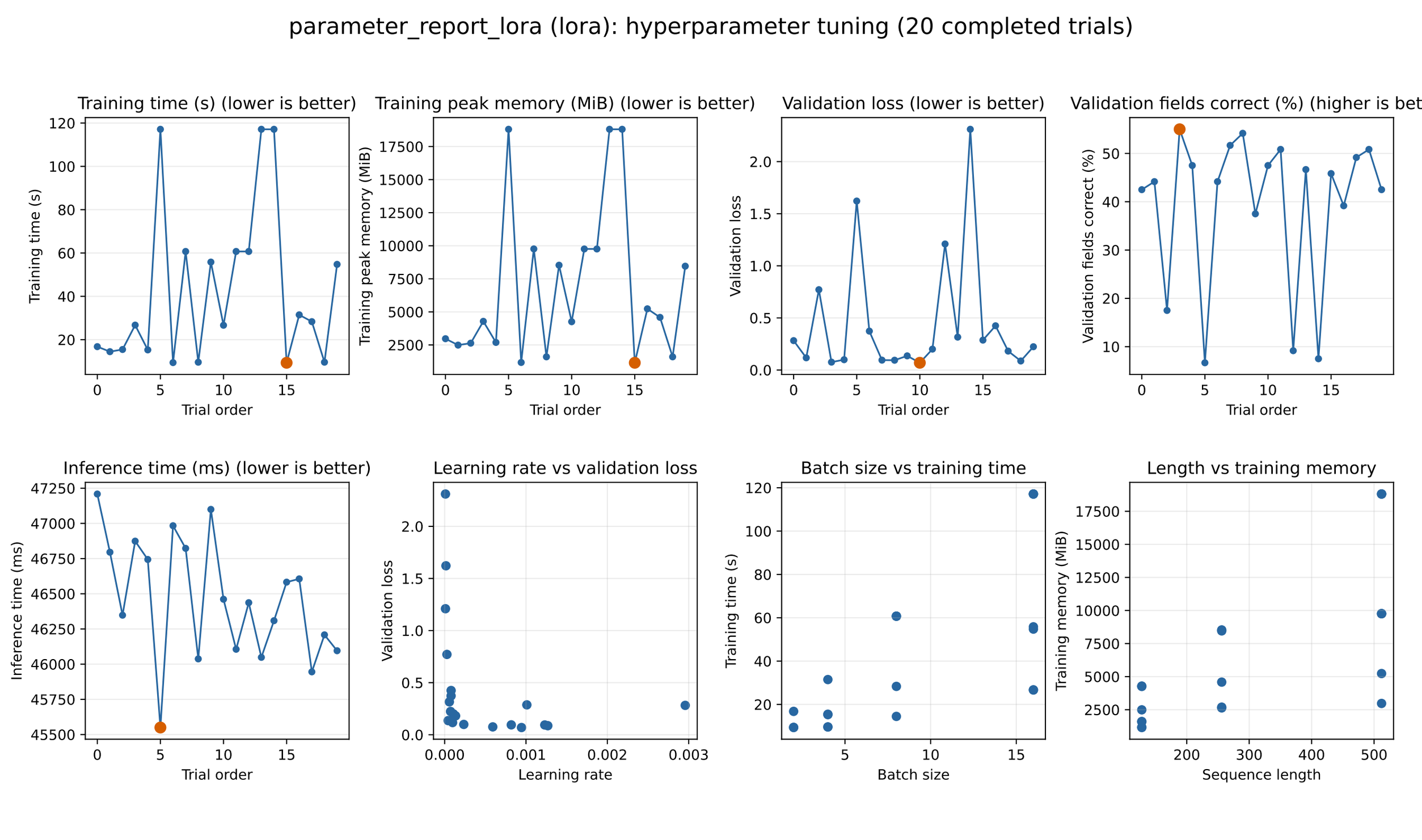




parameter_report_kvcache (kvcache): trial hyperparameters

Trial	Learning Rate	Batch Size	Length
0	0.0029578	2	512
1	3.5613e-05	8	256
2	0.00015536	4	256
3	0.00067814	16	128
4	0.00022618	4	128
5	5.6412e-05	16	512
6	3.3238e-05	16	128
7	0.00016224	16	128
8	8.5841e-05	16	128
9	0.00023393	16	128
10	5.7584e-05	16	128
11	**1.3713e-05**	**16**	**128**
12	1.3997e-05	16	512
13	0.00012607	16	128
14	3.8167e-05	16	512
15	1.4167e-05	16	256
16	1e-05	16	256
17	3.0273e-05	16	256
18	1.6997e-05	16	256
19	9.2127e-05	16	512

Highlighted row: selected best trial based on this report's objectives.


parameter_report_lora (lora): hyperparameter tuning (20 completed trials)
Training time (s) (lower is better)
Training time (s)
Trial order
Training peak memory (MiB) (lower is better)
Training peak memory (MiB)
Trial order
Validation loss (lower is better)
Validation loss
Trial order
Validation fields correct (%) (higher is bet
Validation fields correct (%)
Trial order
Inference time (ms) (lower is better)
Inference time (ms)
Trial order
Learning rate vs validation loss
Validation loss
Learning rate
Batch size vs training time
Training time (s)
Batch size
Length vs training memory
Training memory (MiB)
Sequence length

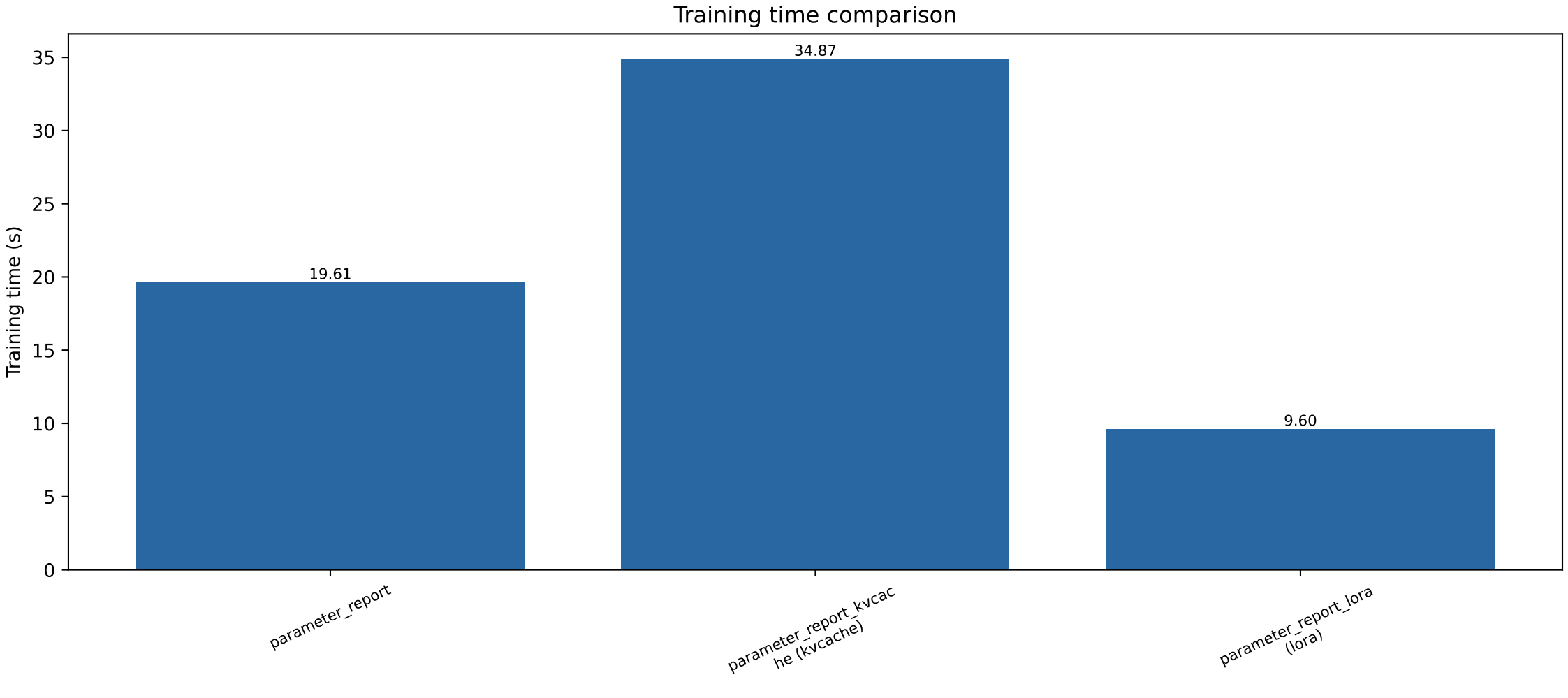
parameter_report_lora (lora): trial hyperparameters

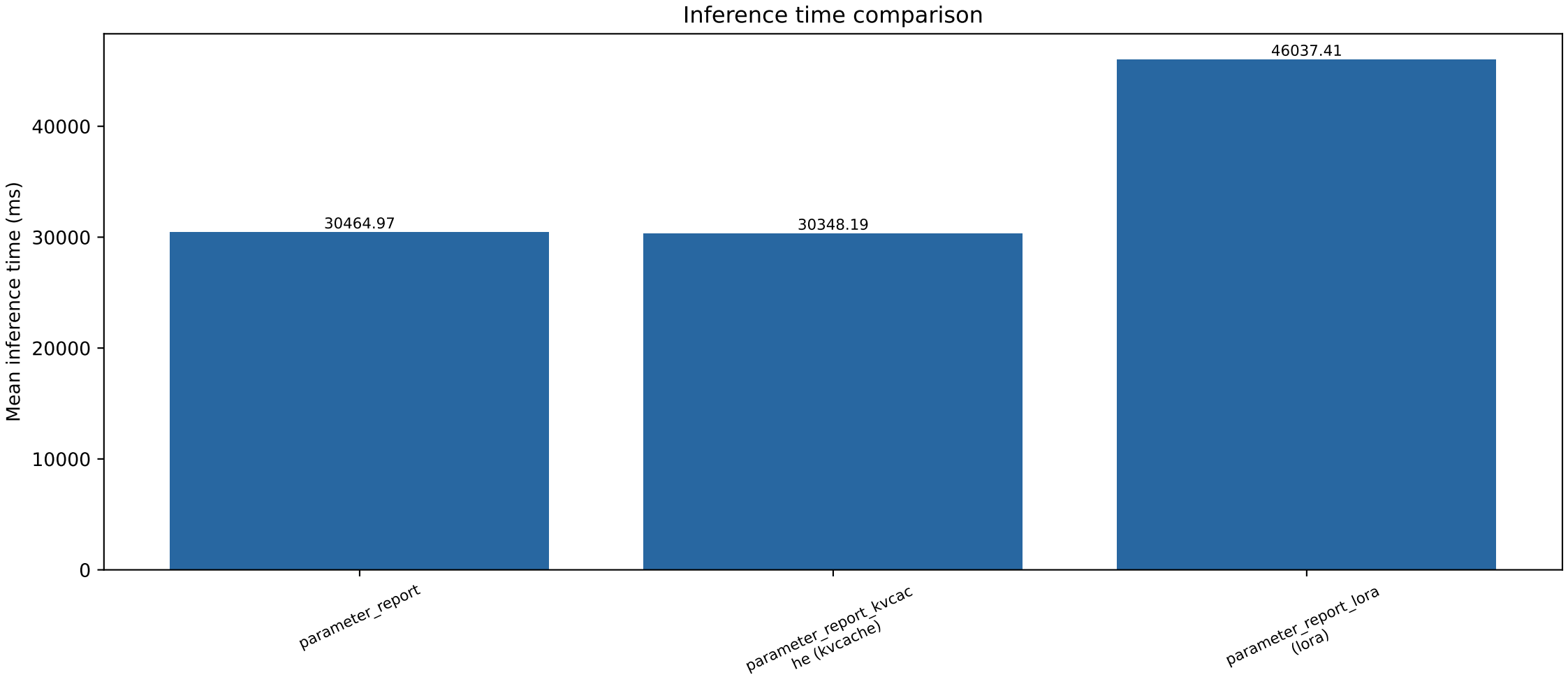
Trial	Learning Rate	Batch Size	Length	Rank	Alpha
0	0.0029578	2	512	8	16
1	9.7448e-05	8	128	8	16
2	2.807e-05	4	256	2	8
3	0.00059155	16	128	16	32
4	0.00023533	4	256	32	64
5	1.5861e-05	16	512	2	4
6	7.9071e-05	2	128	16	8
7	0.00081909	8	512	4	32
8	**0.0012308**	**4**	**128**	**2**	**8**
9	4.2976e-05	16	256	32	32
10	0.00094482	16	128	2	4
11	0.0001072	8	512	2	4
12	1e-05	8	512	2	16
13	5.8035e-05	16	512	2	4
14	1e-05	16	512	2	4
15	0.0010106	2	128	2	64
16	7.953e-05	4	512	2	4
17	0.00013428	8	256	2	4
18	0.0012681	4	128	2	4
19	7.2631e-05	16	256	2	4

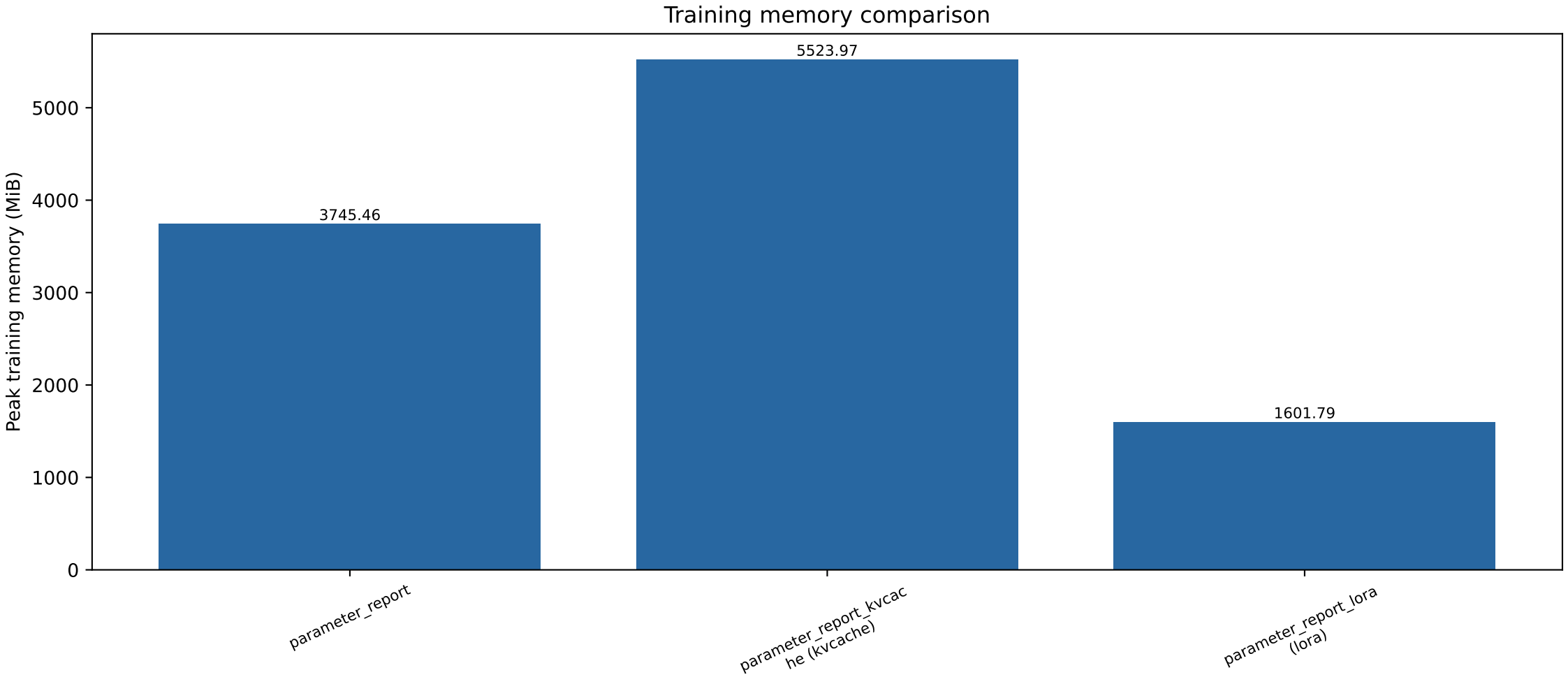
Highlighted row: selected best trial based on this report's objectives.

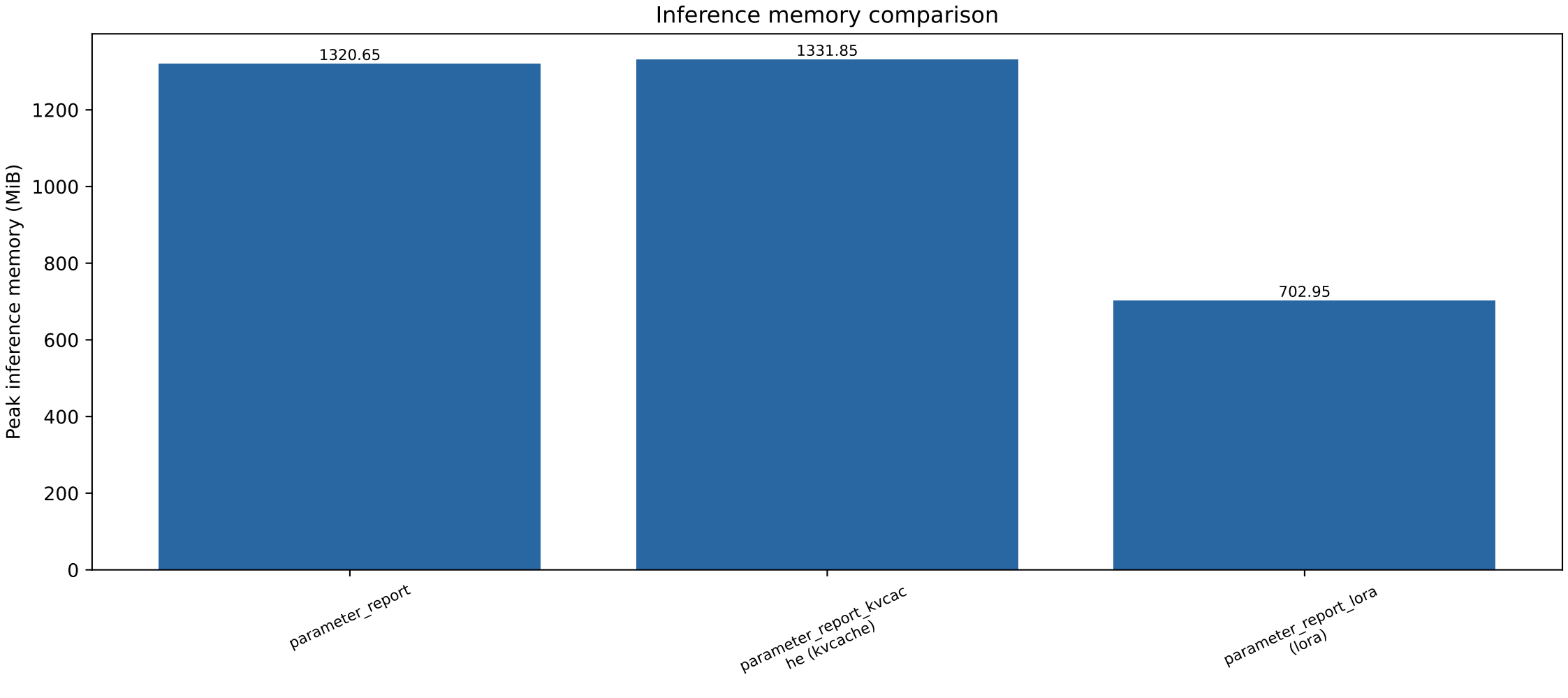
Best trial selected from each parameter-search report

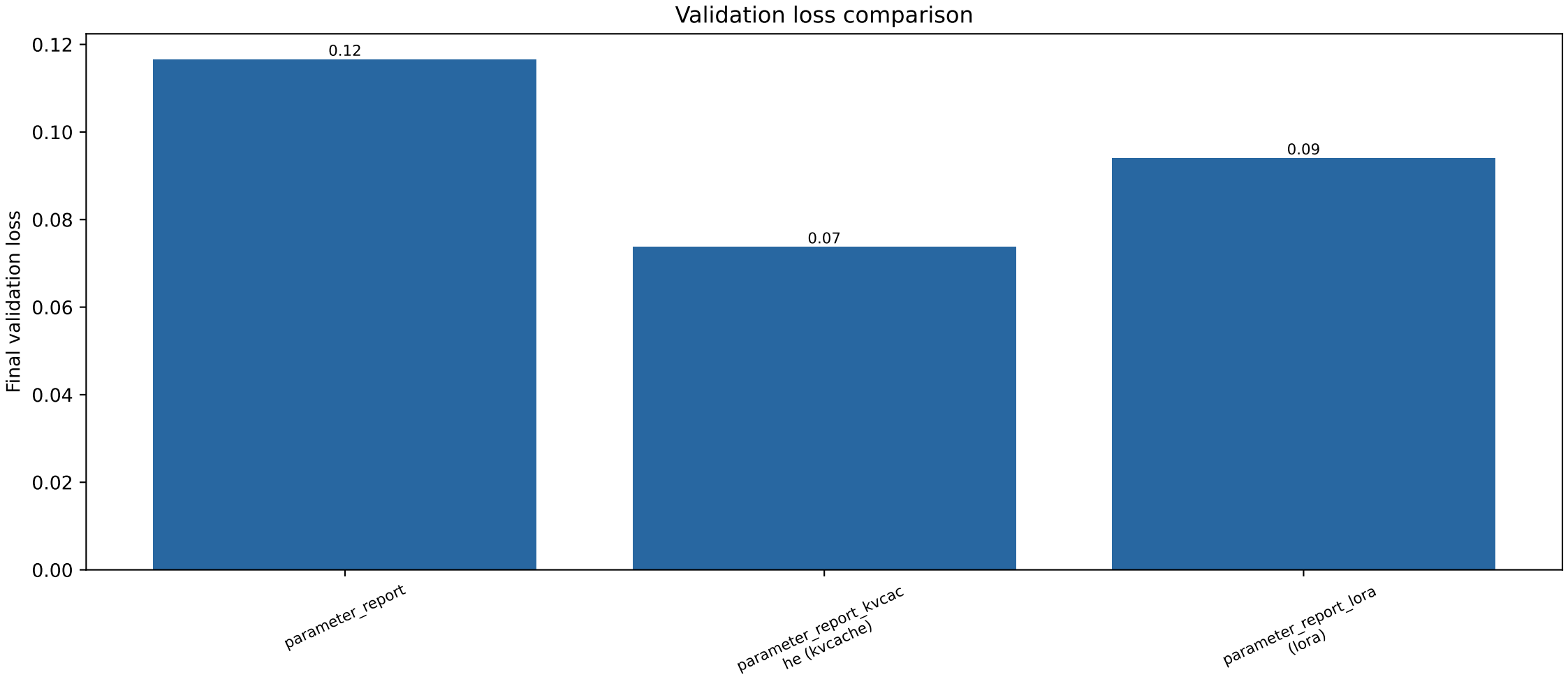
Configuration	Trial	LR	Batch	Length	Rank	Alpha	Train s	Train MiB	Infer ms	Val loss	Val fields	Test fields
parameter_report	7	0.000252	8	128	-	-	19.6	3745	30465.0	0.1166	59.2%	16.7%
parameter_report_kvcache (kvcache)	11	1.371e-05	16	128	-	-	34.9	5524	30348.2	0.07375	50.8%	51.7%
parameter_report_lora (lora)	8	0.001231	4	128	2	8.0	9.6	1602	46037.4	0.09406	54.2%	56.7%

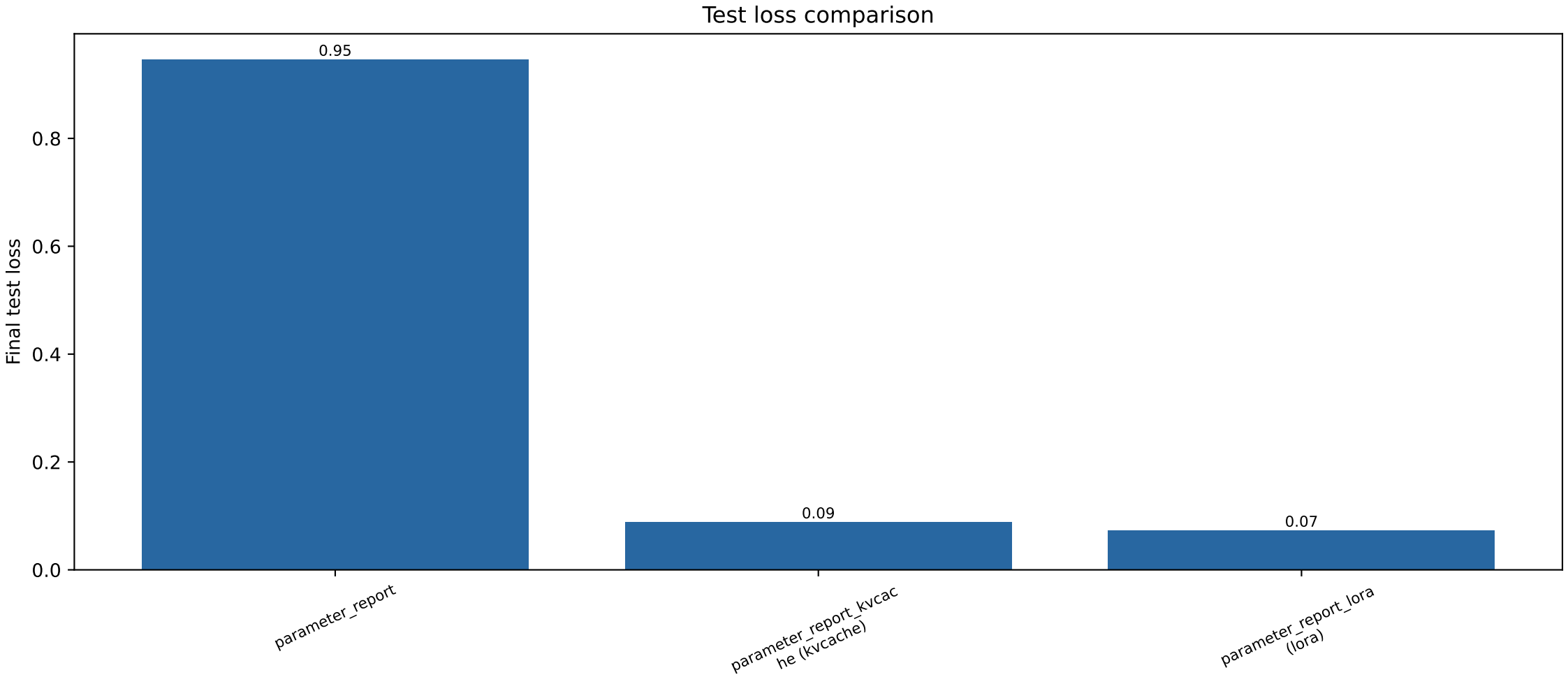

Training time comparison
Training time (s)
0
5
10
15
20
25
30
35
19.61
34.87
9.60
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)

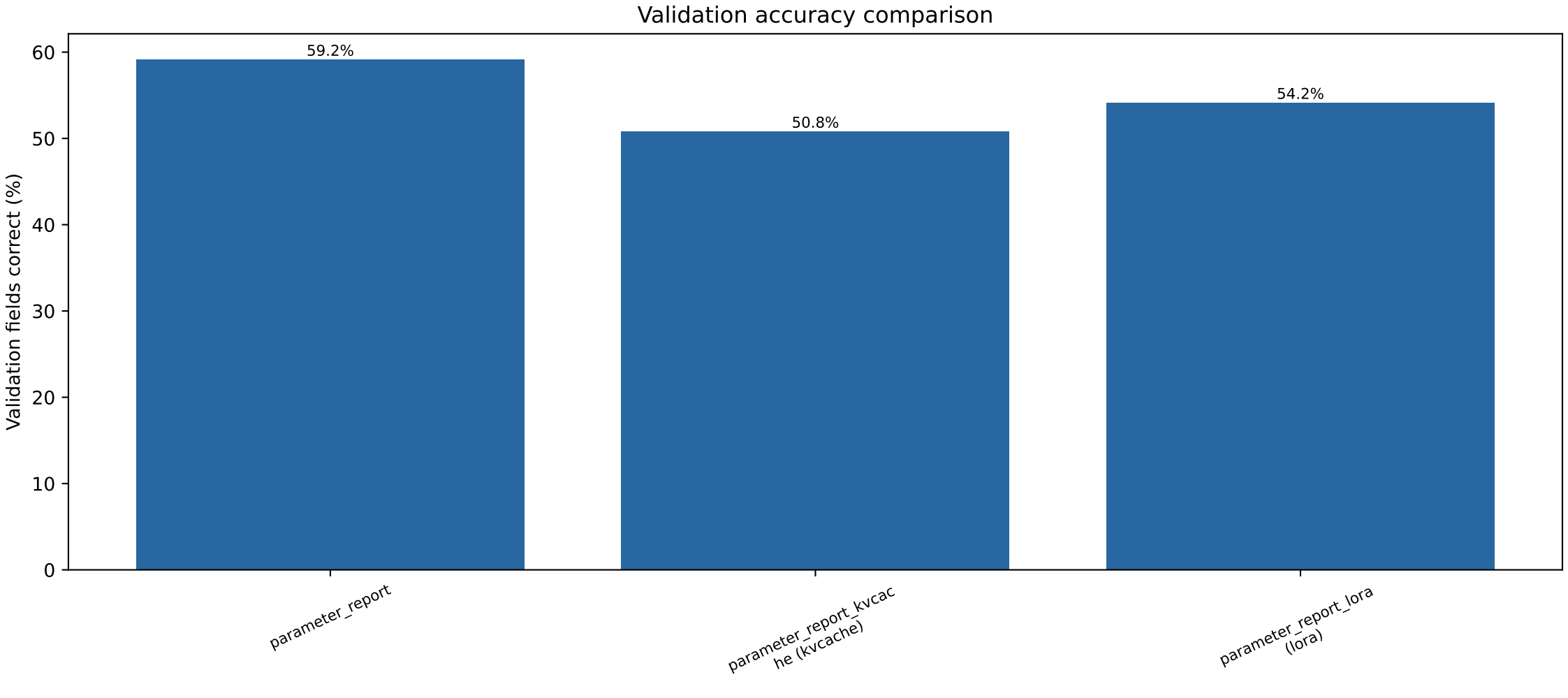

Inference time comparison
Mean inference time (ms)
0
10000
20000
30000
40000
30464.97
30348.19
46037.41
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)

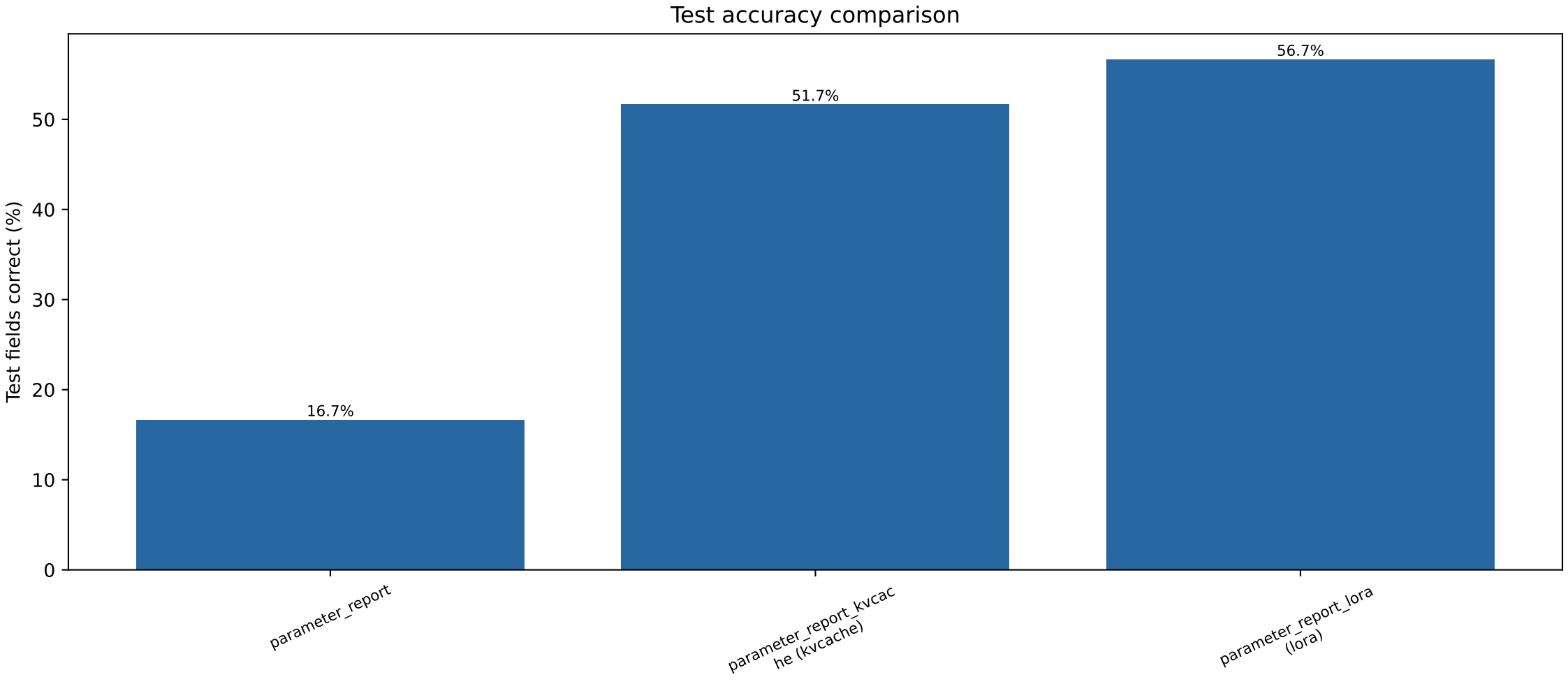

Training memory comparison
Peak training memory (MiB)
0
1000
2000
3000
4000
5000
3745.46
5523.97
1601.79
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)


Inference memory comparison
Peak inference memory (MiB)
0
200
400
600
800
1000
1200
1320.65
1331.85
702.95
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)


Validation loss comparison
Final validation loss
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.12
0.07
0.09
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)


Test loss comparison
Final test loss
0.0
0.2
0.4
0.6
0.8
0.95
0.09
0.07
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)


Validation accuracy comparison
Validation fields correct (%)
0
10
20
30
40
50
60
59.2%
50.8%
54.2%
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)


Test accuracy comparison
Test fields correct (%)
0
10
20
30
40
50
16.7%
51.7%
56.7%
parameter_report
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)